

Association between betel-nut chewing and chronic kidney disease in men.

BACKGROUND: Betel-nut use is associated with metabolic syndrome and obesity. However, the association between betel-nut chewing and risk for chronic kidney disease (CKD) is unknown. The present study was conducted to determine the association between betel-nut chewing and CKD in men. **METHODS:** We retrospectively reviewed health-check records of 3264 men in a hospital-based cross-sectional screening programme from 2003 to 2006. CKD was defined as estimated glomerular filtration rate less than 60 ml/min/1.73 m² calculated by the Modification of Diet in Renal Disease formula. Risk factors for CKD including diabetes, hypertension, BMI, smoking, alcohol consumption and age were also considered. **RESULTS:** A total of 677 (20.7 %) men were found to have CKD and 427 (13.1 %) participants reported a history of betel-nut use. The prevalence (24.8 %) of CKD in betel-nut users was significantly higher than that (11.3 %) of participants without betel-nut use ($P = 0.026$). In multivariate logistic regression analysis with adjustments for age, hypertension, diabetes and hyperlipidaemia, betel-nut use was independently associated with CKD ($P < 0.001$). The adjusted odds ratio for betel-nut use was 2.572 (95 % CI 1.917, 3.451). **CONCLUSIONS:** Betel-nut use is associated with CKD in men. The association between betel-nut use and CKD is independent of age, BMI, smoking, alcohol consumption, hypertension, diabetes and hyperlipidaemia.